PRD – Backend Zone & Category Validation

BoltPe Insurance Service | Backend Validation Only

1. Objective

The backend receives numeric codes for Health Insurance zone and Personal Accident category. For both fields, the backend must accept only 1, 2, or 3. Any other input must be rejected before the downstream insurance API is called.

2. Scope

Insurance Type	Field	Allowed Values
HEALTH	zone	1, 2, 3
PERSONAL_ACCIDENT	category	1, 2, 3

3. Health Insurance – Zone Validation

When insuranceType = HEALTH, zone is mandatory and must contain exactly one supported value: 1, 2, or 3.

Scenario	Input	Expected Result	Backend Message
Valid Zone	1	Valid	-
Valid Zone	2	Valid	-
Valid Zone	3	Valid	-
Missing	Field not provided	Invalid	zone is required
Null	null	Invalid	zone is required
Empty	""	Invalid	zone is required
Blank	" "	Invalid	zone is required
Zero	0	Invalid	zone must be one of: 1, 2, 3
Greater than 3	4, 5, 10	Invalid	zone must be one of: 1, 2, 3
Negative	-1, -2	Invalid	zone must be one of: 1, 2, 3
Decimal	1.5, 2.5	Invalid	zone must be one of: 1, 2, 3
Alphabet	A, abc	Invalid	zone must be one of: 1, 2, 3
Special Character	@, #, %	Invalid	zone must be one of: 1, 2, 3
Alphanumeric	1A, A1	Invalid	zone must be one of: 1, 2, 3
Number + Special Character	1@, 2#	Invalid	zone must be one of: 1, 2, 3
Multiple Digits	12, 23, 123	Invalid	zone must be one of: 1, 2, 3

4. Personal Accident – Category Validation

When insuranceType = PERSONAL_ACCIDENT, category is mandatory and must contain exactly one supported value: 1, 2, or 3.

Scenario	Input	Expected Result	Backend Message
Valid Category	1	Valid	-
Valid Category	2	Valid	-
Valid Category	3	Valid	-
Missing	Field not provided	Invalid	category is required
Null	null	Invalid	category is required
Empty	""	Invalid	category is required
Blank	" "	Invalid	category is required
Zero	0	Invalid	category must be one of: 1, 2, 3
Greater than 3	4, 5, 10	Invalid	category must be one of: 1, 2, 3
Negative	-1, -2	Invalid	category must be one of: 1, 2, 3
Decimal	1.5, 2.5	Invalid	category must be one of: 1, 2, 3
Alphabet	A, abc	Invalid	category must be one of: 1, 2, 3
Special Character	@, #, %	Invalid	category must be one of: 1, 2, 3
Alphanumeric	1A, A1	Invalid	category must be one of: 1, 2, 3
Number + Special Character	1@, 2#	Invalid	category must be one of: 1, 2, 3
Multiple Digits	12, 23, 123	Invalid	category must be one of: 1, 2, 3

5. Backend Business Rules

- BR-01: For HEALTH, zone must be present and zone ∈ {1, 2, 3}.
- BR-02: For PERSONAL_ACCIDENT, category must be present and category ∈ {1, 2, 3}.
- BR-03: Null, missing, empty, and blank values must be rejected.
- BR-04: Alphabets, special characters, alphanumeric values, decimals, negative values, multiple-digit values, and all numbers outside 1–3 must be rejected.
- BR-05: If validation fails, the downstream insurance provider API must not be called.

6. Recommended Backend Validation

If zone/category are String fields, the following validation provides controlled error messages:

```
@NotBlank(message = "zone is required")

@Pattern(regexp = "^[1-3]$", message = "zone must be one of: 1, 2, 3")

private String zone;
```

```
@NotBlank(message = "category is required")

@Pattern(regexp = "^[1-3]$", message = "category must be one of: 1, 2, 3")

private String category;
```

7. Acceptance Criteria

- AC-01: HEALTH zone values 1, 2, and 3 must be accepted.
- AC-02: Any HEALTH zone value other than 1, 2, or 3 must be rejected.
- AC-03: PERSONAL_ACCIDENT category values 1, 2, and 3 must be accepted.
- AC-04: Any PERSONAL_ACCIDENT category value other than 1, 2, or 3 must be rejected.
- AC-05: Missing, null, empty, and blank values must be rejected.
- AC-06: Alphabets and special characters must be rejected.
- AC-07: Decimal and negative values must be rejected.
- AC-08: Multiple-digit values such as 12, 23, and 123 must be rejected.
- AC-09: On validation failure, downstream insurance processing must not start.

8. Final Validation Summary

Insurance Type	Field	Backend Accepts	Everything Else
HEALTH	zone	Only 1, 2, 3	Invalid
PERSONAL_ACCIDENT	category	Only 1, 2, 3	Invalid

Final Rule: Allowed = {1, 2, 3}. Everything else is invalid. Frontend mapping and display values are outside the scope of this requirement.